Altitudes, medians, and bisectors of triangles

Core Concepts: Altitudes, medians, and bisectors of triangles

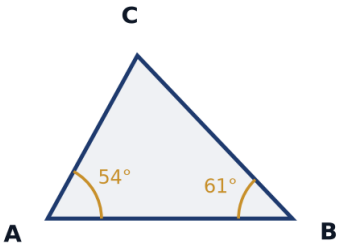
1 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 93°

b. 103°

c. 85°

d. 138°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 10

c. 8

d. 11

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 86°

b. 96°

c. 78°

d. 138°

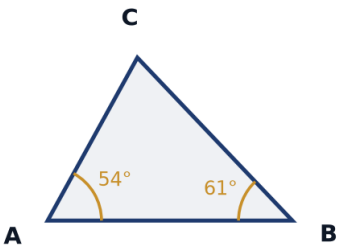
5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 8

b. 13

c. 11

d. 14

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 79°

b. 89°

c. 71°

d. 138°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

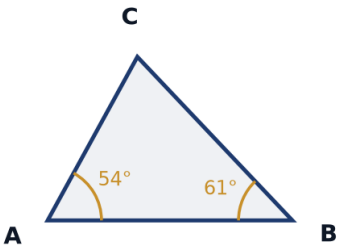
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?

a. 11

b. 16

c. 14

d. 17



Exam-Style Problem Solving & Modeling

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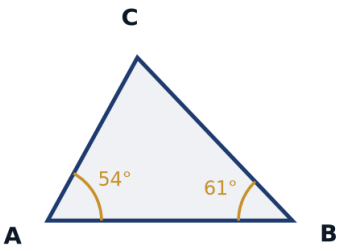
10 In $\triangle ABC$ studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 72°

b. 82°

c. 64°

d. 138°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 14

b. 19

c. 17

d. 20

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